classification-torch

January 28, 2026

1 Feature Space and Classification Lab

The objectives of this lab are for you to … 1. Conceptualize the transition from the image space to feature space 1. Visualize the feature space and to 1. Understand manifolds and discriminability in feature space 1. Experiment on this feature space with simple classifiers 1. Assess the performance of the classifiers for visual recognition

1.1 Libraries, data set, sanity check

[1]:
```
import sys
print(sys.executable)

import torch
print("torch:", torch.__version__)

import cv2
print("cv2:", cv2.__version__)
```

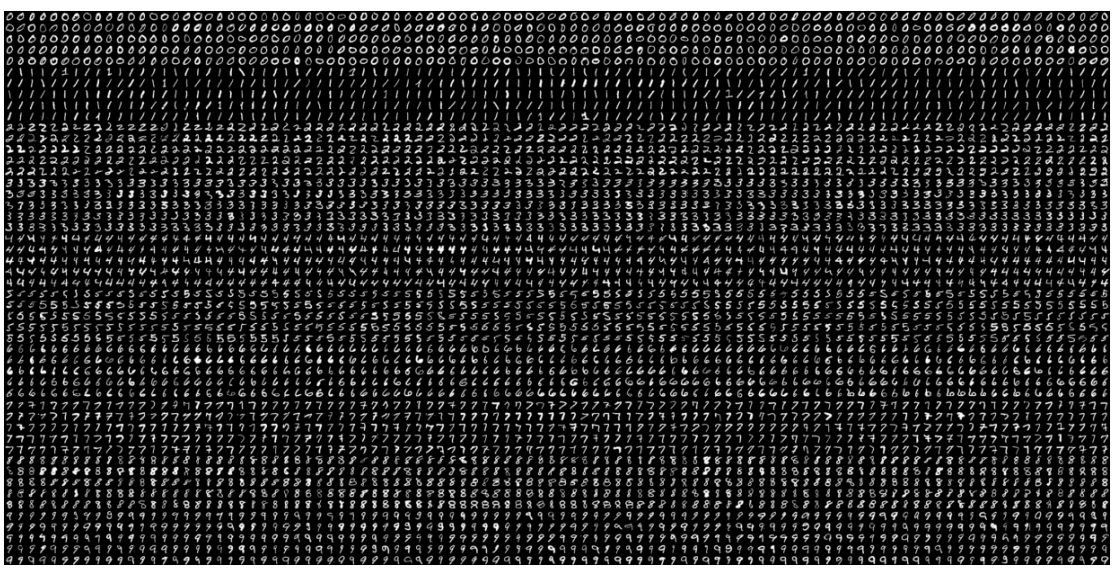
```
c:\Users\jgooh\anaconda3\envs\cs4330\python.exe
torch: 2.2.2
cv2: 4.10.0
```

[2]:
```
# one-time prep work
# set matplotlib to render in this window, with the specified figure size
%matplotlib inline
import random
import numpy as np
import cv2 as cv
print(f"Your OpenCV Version is {cv.__version__} - please review the␣
↪documentation: "
      f"https://docs.opencv.org/{cv.__version__}/")
import matplotlib.pyplot as plt
```

```
Your OpenCV Version is 4.10.0 - please review the documentation:
https://docs.opencv.org/4.10.0/
```

The code and data for this this lab come from various key sources, including scikit-learn and OpenCV. Please spend 10 minutes to look through the OpenCV Documentation (select the version that matches what you have installed) to get an overview of the available functionality.

[3]:
```
# Load digits.png: this single large image of 50x100 digits, each 20x20 in␣
↪size,
# used to be in OpenCV/data. You can find it on the web, search for "opencv␣
↪digits.png".
img = cv.imread('digits.png')
plt.figure(figsize=(16, 8));
plt.gray()
plt.imshow(img), plt.axis('off'), plt.show();

# if you cannot find this image, you can alternatively work with a built-in␣
↪dataset,
# but you will need to convert the cell sizing, train and test sets, etc.
# See here for usage info: https://scikit-learn.org/stable/auto_examples/
↪manifold/plot_lle_digits.html
#from sklearn.datasets import load_digits
#digits = load_digits()
#print(digits.data.shape) # 10 classes, about 180 images/class, 8x8 images

gray = cv.cvtColor(img,cv.COLOR_BGR2GRAY)
# Now we split the image to 5000 cells, each 20x20 size
cells = [np.hsplit(row,100) for row in np.vsplit(gray,50)]
# Make it into a Numpy array: its size will be (50,100,20,20).
# The first 5 rows are of 0s, the next of 1s, and so on.
x = np.array(cells)
```



[4]:
```
# Prepare the training data and test data including labels
train = x[:,:50].reshape(-1,400).astype(np.float32) # Size = (2500,400) with␣
↪400=20x20 pixels
```

```
test = x[:,50:100].reshape(-1,400).astype(np.float32) # Size = (2500,400)
# Create labels for train and test data
k = np.arange(10)
train_labels = np.repeat(k,250)[:,np.newaxis]
test_labels = train_labels.copy()

# Sanity check: render random digits and their labels. There are 250 of each␣
↪class.
for i in np.arange(10):
    plt.subplot(1, 10, i+1)
    r = random.randint(i*250,(i+1)*250-1)
    ri = train[r,:].reshape(20, 20)
    plt.imshow(ri), plt.title(train_labels[r,0]), plt.axis('off')
plt.show();
```

1.2 Neural Networks

[5]:
```
import torch
from torch import nn
from torch.utils.data import TensorDataset, DataLoader
from torchvision import datasets
from torchvision.transforms import ToTensor
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
```

[6]:
```
print("type(train):", type(train))
try:
    print("train dtype:", getattr(train, "dtype", None))
    print("train shape:", getattr(train, "shape", None))
except Exception as e:
    print("error inspecting train:", e)

print("first element type:", type(train[0]) if hasattr(train, "__len__") else␣
↪None)
print("train[0]:", train[0] if hasattr(train, "__len__") else None)
```

```
type(train): <class 'numpy.ndarray'>
train dtype: float32
train shape: (2500, 400)
first element type: <class 'numpy.ndarray'>
```

```
train[0]: [  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   9.  33.   9.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
  41. 177. 249. 178.  29.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.  33. 198. 255. 240. 255. 107.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   1.  70. 199. 255. 255.
 197. 154. 253.  98.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.  45. 238. 255. 205. 224. 222.  83. 224. 128.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.  25. 202. 255. 193.  40.  99.  54.   0.
 190. 197.  16.   0.   0.   0.   0.   0.   0.   0.   0.  20. 163. 246.
 152.  72.   0.   0.   0.   0. 184. 252.  74.   0.   0.   0.   0.   0.
   0.   0.   0.  97. 255. 118.   0.   1.   0.   0.   0.   0. 184. 255.
  82.   0.   0.   0.   0.   0.   0.   0.  20. 218. 216.  17.   0.   0.
   0.   0.   0.   0. 183. 255.  78.   0.   0.   0.   0.   0.   0.   0.
  67. 255. 138.   0.   0.   0.   0.   0.   0.  24. 215. 188.  22.   0.
   0.   0.   0.   0.   0.   0.  69. 247.  87.   0.   0.   0.   0.   0.
  31. 185. 212.  54.   0.   0.   0.   0.   0.   0.   0.   0.  71. 237.
  68.   0.   0.   0.   0.  94. 211. 177.  17.   0.   0.   0.   0.   0.
   0.   0.   0.   0.  69. 255. 170.  39.  39. 115. 183. 238. 185.  37.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  57. 245. 255. 229.
 233. 255. 216. 117.  13.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   7. 114. 230. 255. 234. 137.  38.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  26.  38.  27.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.]
```

[7]:

```python
# Convert numpy arrays to PyTorch tensors
train_tensor = torch.tensor(train)
train_labels_tensor = torch.tensor(train_labels).squeeze().long()
test_tensor = torch.tensor(test)
test_labels_tensor  = torch.tensor(test_labels).squeeze().long()

# Create datasets
train_dataset = TensorDataset(train_tensor, train_labels_tensor)
test_dataset = TensorDataset(test_tensor, test_labels_tensor)

# Create data loaders
batch_size = 64
train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

# Print dataset sizes
print(f"Training set size: {len(train_dataset)} samples")
```

```python
print(f"Test set size: {len(test_dataset)} samples")
print(f"Number of batches in train_loader: {len(train_loader)}")
print(f"Number of batches in test_loader: {len(test_loader)}")

# Show an example batch
sample_batch = next(iter(train_loader))
print(f"\nSample batch - Features shape: {sample_batch[0].shape}, Labels shape:␣
↪{sample_batch[1].shape}")
```

```
Training set size: 2500 samples
Test set size: 2500 samples
Number of batches in train_loader: 40
Number of batches in test_loader: 40

Sample batch - Features shape: torch.Size([64, 400]), Labels shape:
torch.Size([64])
```

[8]:
```python
# Get cpu or gpu device for training.
device = "cuda" if torch.cuda.is_available() else "mps" if torch.backends.mps.
↪is_available() else "cpu"
print(f"Using {device} device")

# Define model
class NeuralNetwork(nn.Module):
    def __init__(self):
        super().__init__()
        self.flatten = nn.Flatten()
        self.linear_relu_stack = nn.Sequential(
            nn.Linear(20*20, 64),
            nn.ReLU(),
            nn.Linear(64, 64),
            nn.ReLU(),
            nn.Linear(64, 10)
        )

    def forward(self, x):
        x = self.flatten(x)
        logits = self.linear_relu_stack(x)
        return logits

model = NeuralNetwork().to(device)
print(model)
```

```
Using cpu device
NeuralNetwork(
  (flatten): Flatten(start_dim=1, end_dim=-1)
  (linear_relu_stack): Sequential(
    (0): Linear(in_features=400, out_features=64, bias=True)
```

```
    (1): ReLU()
    (2): Linear(in_features=64, out_features=64, bias=True)
    (3): ReLU()
    (4): Linear(in_features=64, out_features=10, bias=True)
  )
)
```

[9]:
```
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(model.parameters(), lr=1e-3)

def train_epoch(dataloader, model, loss_fn, optimizer):
    size = len(dataloader.dataset)
    model.train()
    for batch, (X, y) in enumerate(dataloader):
        X, y = X.to(device), y.to(device)

        # Compute prediction error
        pred = model(X)
        loss = loss_fn(pred, y)

        # Backpropagation
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        if batch % 100 == 0:
            loss, current = loss.item(), batch * len(X)
            print(f"loss: {loss:>7f}  [{current:>5d}/{size:>5d}]")

def test(dataloader, model, loss_fn):
    size = len(dataloader.dataset)
    num_batches = len(dataloader)
    model.eval()
    test_loss, correct = 0, 0
    with torch.no_grad():
        for X, y in dataloader:
            X, y = X.to(device), y.to(device)
            pred = model(X)
            test_loss += loss_fn(pred, y).item()
            correct += (pred.argmax(1) == y).type(torch.float).sum().item()
    test_loss /= num_batches
    correct /= size
    print(f"Test Error: \n Accuracy: {(100*correct):>0.1f}%, Avg loss:␣
 ↪{test_loss:>8f} \n")
```

[10]:
```
epochs = 100
for t in range(epochs):
```

```python
    print(f"Epoch {t+1}\n-------------------------------")
    train_epoch(train_loader, model, loss_fn, optimizer)
    test(test_loader, model, loss_fn)
print("Done!")
```

```
Epoch 1
-------------------------------
loss: 19.182579  [    0/ 2500]
Test Error:
 Accuracy: 65.9%, Avg loss: 1.222933

Epoch 2
-------------------------------
loss: 1.117409  [    0/ 2500]
Test Error:
 Accuracy: 64.5%, Avg loss: 1.396089

Epoch 3
-------------------------------
loss: 1.050106  [    0/ 2500]
Test Error:
 Accuracy: 72.2%, Avg loss: 1.008489

Epoch 4
-------------------------------
loss: 0.694618  [    0/ 2500]
Test Error:
 Accuracy: 79.4%, Avg loss: 0.785004

Epoch 5
-------------------------------
loss: 0.521854  [    0/ 2500]
Test Error:
 Accuracy: 81.3%, Avg loss: 0.710710

Epoch 6
-------------------------------
loss: 0.537980  [    0/ 2500]
Test Error:
 Accuracy: 80.1%, Avg loss: 0.743495

Epoch 7
-------------------------------
loss: 0.340361  [    0/ 2500]
Test Error:
 Accuracy: 81.8%, Avg loss: 0.702622

Epoch 8
```

```
-------------------------------
loss: 0.471806  [    0/ 2500]
Test Error:
 Accuracy: 85.9%, Avg loss: 0.583508

Epoch 9
-------------------------------
loss: 0.322507  [    0/ 2500]
Test Error:
 Accuracy: 82.9%, Avg loss: 0.768900

Epoch 10
-------------------------------
loss: 0.191606  [    0/ 2500]
Test Error:
 Accuracy: 85.2%, Avg loss: 0.611932

Epoch 11
-------------------------------
loss: 0.224932  [    0/ 2500]
Test Error:
 Accuracy: 86.3%, Avg loss: 0.545763

Epoch 12
-------------------------------
loss: 0.155913  [    0/ 2500]
Test Error:
 Accuracy: 86.8%, Avg loss: 0.564110

Epoch 13
-------------------------------
loss: 0.178046  [    0/ 2500]
Test Error:
 Accuracy: 85.8%, Avg loss: 0.579533

Epoch 14
-------------------------------
loss: 0.189328  [    0/ 2500]
Test Error:
 Accuracy: 87.4%, Avg loss: 0.530298

Epoch 15
-------------------------------
loss: 0.222569  [    0/ 2500]
Test Error:
 Accuracy: 87.5%, Avg loss: 0.517023

Epoch 16
```

```
-------------------------------
loss: 0.058975  [    0/ 2500]
Test Error:
 Accuracy: 87.2%, Avg loss: 0.538783

Epoch 17
-------------------------------
loss: 0.068778  [    0/ 2500]
Test Error:
 Accuracy: 86.3%, Avg loss: 0.555735

Epoch 18
-------------------------------
loss: 0.235269  [    0/ 2500]
Test Error:
 Accuracy: 71.0%, Avg loss: 1.510414

Epoch 19
-------------------------------
loss: 1.047158  [    0/ 2500]
Test Error:
 Accuracy: 87.1%, Avg loss: 0.538902

Epoch 20
-------------------------------
loss: 0.195758  [    0/ 2500]
Test Error:
 Accuracy: 87.9%, Avg loss: 0.511619

Epoch 21
-------------------------------
loss: 0.061861  [    0/ 2500]
Test Error:
 Accuracy: 86.8%, Avg loss: 0.527790

Epoch 22
-------------------------------
loss: 0.197849  [    0/ 2500]
Test Error:
 Accuracy: 87.9%, Avg loss: 0.516384

Epoch 23
-------------------------------
loss: 0.207035  [    0/ 2500]
Test Error:
 Accuracy: 88.2%, Avg loss: 0.505809

Epoch 24
```

```
-------------------------------
loss: 0.126066  [    0/ 2500]
Test Error:
 Accuracy: 71.0%, Avg loss: 1.601959

Epoch 25
-------------------------------
loss: 0.949211  [    0/ 2500]
Test Error:
 Accuracy: 87.6%, Avg loss: 0.518075

Epoch 26
-------------------------------
loss: 0.125548  [    0/ 2500]
Test Error:
 Accuracy: 87.8%, Avg loss: 0.523181

Epoch 27
-------------------------------
loss: 0.115113  [    0/ 2500]
Test Error:
 Accuracy: 88.3%, Avg loss: 0.508915

Epoch 28
-------------------------------
loss: 0.014596  [    0/ 2500]
Test Error:
 Accuracy: 88.3%, Avg loss: 0.533301

Epoch 29
-------------------------------
loss: 0.048467  [    0/ 2500]
Test Error:
 Accuracy: 88.4%, Avg loss: 0.508503

Epoch 30
-------------------------------
loss: 0.028845  [    0/ 2500]
Test Error:
 Accuracy: 88.4%, Avg loss: 0.512276

Epoch 31
-------------------------------
loss: 0.056458  [    0/ 2500]
Test Error:
 Accuracy: 87.9%, Avg loss: 0.525075

Epoch 32
```

```
-------------------------------
loss: 0.053722  [    0/ 2500]
Test Error:
 Accuracy: 88.1%, Avg loss: 0.516469

Epoch 33
-------------------------------
loss: 0.049560  [    0/ 2500]
Test Error:
 Accuracy: 88.5%, Avg loss: 0.507814

Epoch 34
-------------------------------
loss: 0.033311  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.519617

Epoch 35
-------------------------------
loss: 0.060630  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.511761

Epoch 36
-------------------------------
loss: 0.015476  [    0/ 2500]
Test Error:
 Accuracy: 88.3%, Avg loss: 0.523181

Epoch 37
-------------------------------
loss: 0.072421  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.509454

Epoch 38
-------------------------------
loss: 0.030864  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.508778

Epoch 39
-------------------------------
loss: 0.030873  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.509273

Epoch 40
```

```
-------------------------------
loss: 0.023699  [    0/ 2500]
Test Error:
 Accuracy: 88.4%, Avg loss: 0.514274

Epoch 41
-------------------------------
loss: 0.097207  [    0/ 2500]
Test Error:
 Accuracy: 88.4%, Avg loss: 0.517609

Epoch 42
-------------------------------
loss: 0.012727  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.512468

Epoch 43
-------------------------------
loss: 0.017913  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.513224

Epoch 44
-------------------------------
loss: 0.011946  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.511787

Epoch 45
-------------------------------
loss: 0.032576  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.517174

Epoch 46
-------------------------------
loss: 0.026492  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.517756

Epoch 47
-------------------------------
loss: 0.064474  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.517618

Epoch 48
```

```
-------------------------------
loss: 0.016173  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.518877

Epoch 49
-------------------------------
loss: 0.022243  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.516543

Epoch 50
-------------------------------
loss: 0.014662  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.517513

Epoch 51
-------------------------------
loss: 0.023574  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.520101

Epoch 52
-------------------------------
loss: 0.025510  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.519146

Epoch 53
-------------------------------
loss: 0.013733  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.521058

Epoch 54
-------------------------------
loss: 0.040911  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.521941

Epoch 55
-------------------------------
loss: 0.017971  [    0/ 2500]
Test Error:
 Accuracy: 88.7%, Avg loss: 0.527255

Epoch 56
```

```
-------------------------------
loss: 0.016416  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.519534

Epoch 57
-------------------------------
loss: 0.014228  [    0/ 2500]
Test Error:
 Accuracy: 87.5%, Avg loss: 0.579342

Epoch 58
-------------------------------
loss: 0.071758  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.523425

Epoch 59
-------------------------------
loss: 0.016849  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.531458

Epoch 60
-------------------------------
loss: 0.031174  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.528750

Epoch 61
-------------------------------
loss: 0.014306  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.528791

Epoch 62
-------------------------------
loss: 0.019075  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.534698

Epoch 63
-------------------------------
loss: 0.014764  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.530469

Epoch 64
```

```
-------------------------------
loss: 0.036653  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.533546

Epoch 65
-------------------------------
loss: 0.007228  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.532753

Epoch 66
-------------------------------
loss: 0.009789  [    0/ 2500]
Test Error:
 Accuracy: 89.4%, Avg loss: 0.532932

Epoch 67
-------------------------------
loss: 0.014484  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.533770

Epoch 68
-------------------------------
loss: 0.021063  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.532964

Epoch 69
-------------------------------
loss: 0.008660  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.532847

Epoch 70
-------------------------------
loss: 0.010180  [    0/ 2500]
Test Error:
 Accuracy: 87.6%, Avg loss: 0.587585

Epoch 71
-------------------------------
loss: 0.046647  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.535978

Epoch 72
```

```
-------------------------------
loss: 0.010611  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.536964

Epoch 73
-------------------------------
loss: 0.013651  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.536264

Epoch 74
-------------------------------
loss: 0.010436  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.542052

Epoch 75
-------------------------------
loss: 0.008353  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.542627

Epoch 76
-------------------------------
loss: 0.007122  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.540199

Epoch 77
-------------------------------
loss: 0.011315  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.539612

Epoch 78
-------------------------------
loss: 0.011908  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.541274

Epoch 79
-------------------------------
loss: 0.009201  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.538404

Epoch 80
```

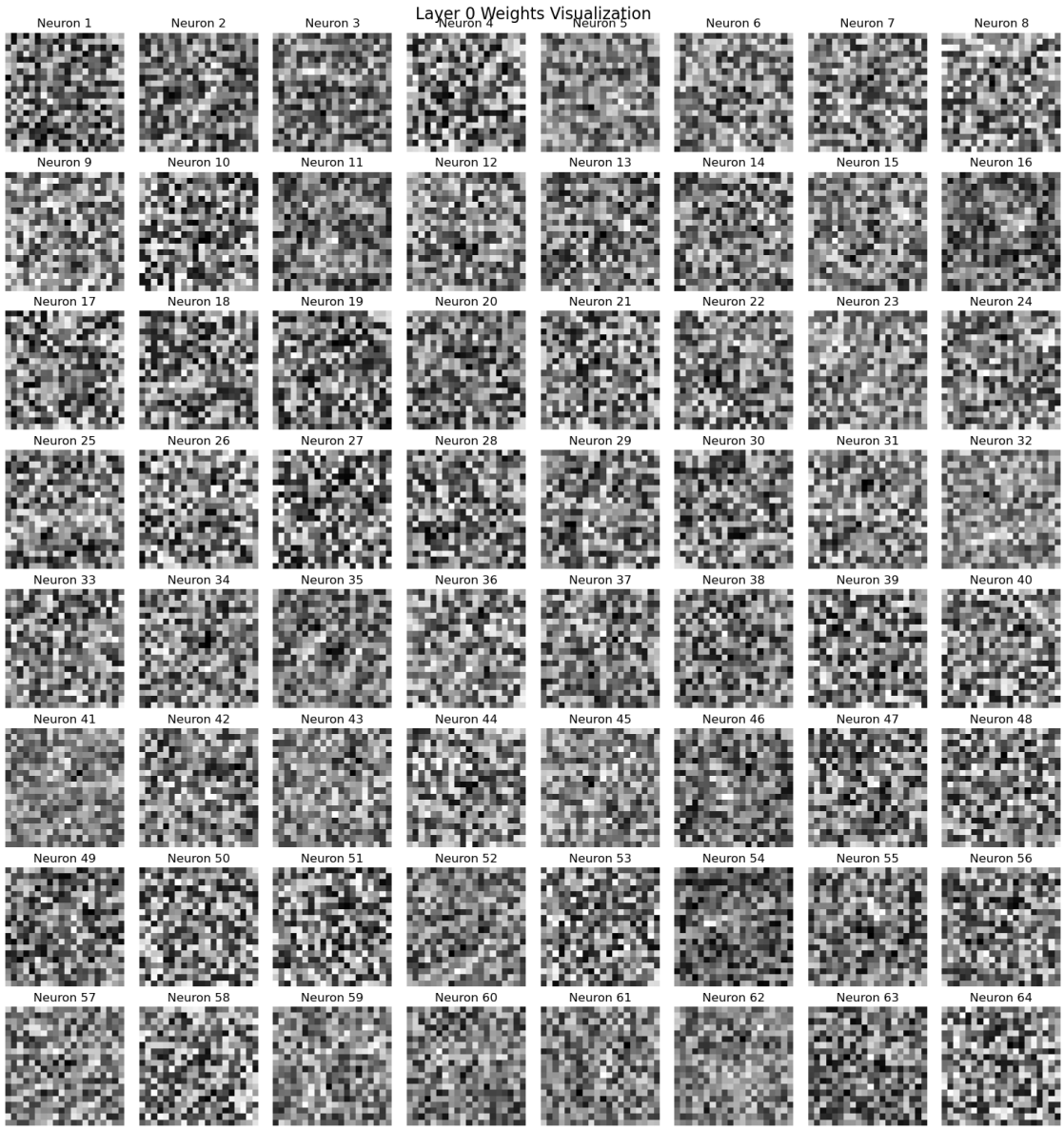
```
-------------------------------
loss: 0.007680  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.540401

Epoch 81
-------------------------------
loss: 0.011568  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.540909

Epoch 82
-------------------------------
loss: 0.007917  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.541127

Epoch 83
-------------------------------
loss: 0.008809  [    0/ 2500]
Test Error:
 Accuracy: 89.0%, Avg loss: 0.541232

Epoch 84
-------------------------------
loss: 0.007150  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.542123

Epoch 85
-------------------------------
loss: 0.016353  [    0/ 2500]
Test Error:
 Accuracy: 88.9%, Avg loss: 0.578003

Epoch 86
-------------------------------
loss: 0.021279  [    0/ 2500]
Test Error:
 Accuracy: 89.4%, Avg loss: 0.542398

Epoch 87
-------------------------------
loss: 0.003208  [    0/ 2500]
Test Error:
 Accuracy: 89.1%, Avg loss: 0.544098

Epoch 88
```

```
-------------------------------
loss: 0.004602  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.543328

Epoch 89
-------------------------------
loss: 0.012350  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.545663

Epoch 90
-------------------------------
loss: 0.009594  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.545431

Epoch 91
-------------------------------
loss: 0.003571  [    0/ 2500]
Test Error:
 Accuracy: 89.2%, Avg loss: 0.544788

Epoch 92
-------------------------------
loss: 0.010716  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.545756

Epoch 93
-------------------------------
loss: 0.008041  [    0/ 2500]
Test Error:
 Accuracy: 89.4%, Avg loss: 0.545552

Epoch 94
-------------------------------
loss: 0.005262  [    0/ 2500]
Test Error:
 Accuracy: 88.8%, Avg loss: 0.547933

Epoch 95
-------------------------------
loss: 0.009741  [    0/ 2500]
Test Error:
 Accuracy: 89.5%, Avg loss: 0.547417

Epoch 96
```

```
-------------------------------
loss: 0.012242  [    0/ 2500]
Test Error:
 Accuracy: 89.4%, Avg loss: 0.547112

Epoch 97
-------------------------------
loss: 0.011272  [    0/ 2500]
Test Error:
 Accuracy: 89.3%, Avg loss: 0.548687

Epoch 98
-------------------------------
loss: 0.005775  [    0/ 2500]
Test Error:
 Accuracy: 89.4%, Avg loss: 0.548007

Epoch 99
-------------------------------
loss: 0.004872  [    0/ 2500]
Test Error:
 Accuracy: 89.5%, Avg loss: 0.548806

Epoch 100
-------------------------------
loss: 0.005643  [    0/ 2500]
Test Error:
 Accuracy: 89.6%, Avg loss: 0.551052

Done!
```

[11]:
```python
def visualize_layer_weights(model, layer_num=0):
    """
    Visualize the weights of a specified layer in the neural network.

    Args:
        model: The PyTorch model
        layer_num: Index of the layer to visualize (0 for first hidden layer, 1␣
↪for second, etc.)
    """
    linear_layers = [module for module in model.modules()
                     if isinstance(module, torch.nn.Linear)]
    if layer_num < 0 or layer_num >= len(linear_layers):
        print(f"Error: Layer number must be between 0 and␣
↪{len(linear_layers)-1}")
        return
    with torch.no_grad():
```

```
        weights = linear_layers[layer_num].weight.cpu().numpy()

    # Determine if we can reshape to square (only for input layer or layers␣
↪with square input)
    input_size = weights.shape[1]
    img_size = int(np.sqrt(input_size))
    can_reshape = (img_size * img_size == input_size)

    # Create a figure to display the weights
    ncols = 8
    nrows = (weights.shape[0] + ncols - 1) // ncols
    plt.figure(figsize=(15, 2 * nrows))
    plt.suptitle(f'Layer {layer_num} Weights Visualization', fontsize=16)

    for i in range(min(weights.shape[0], nrows * ncols)):  # for each subplot
        plt.subplot(nrows, ncols, i + 1)
        if can_reshape:
            weight_img = weights[i].reshape(img_size, img_size)
        else:
            weight_img = weights[i]

        #weight_img = (weight_img - weight_img.min()) / (weight_img.max() -␣
↪weight_img.min() + 1e-6)
        if can_reshape:
            plt.imshow(weight_img, cmap='gray')
        else:
            # For non-square layers, plot as a 1D array
            plt.plot(weight_img)
            plt.axis('off')

        plt.axis('off')
        plt.title(f'Neuron {i+1}')

    plt.tight_layout()
    plt.show()

    # Print layer information
    print(f"Visualizing Layer {layer_num} of shape {weights.shape}")
    print(f"Number of neurons: {weights.shape[0]}")
    print(f"Input features per neuron: {weights.shape[1]}")

visualize_layer_weights(model, layer_num=0) # change from 0..2 to visualize␣
↪each of the 3 layers
```




```
Visualizing Layer 0 of shape (64, 400)
Number of neurons: 64
Input features per neuron: 400
```

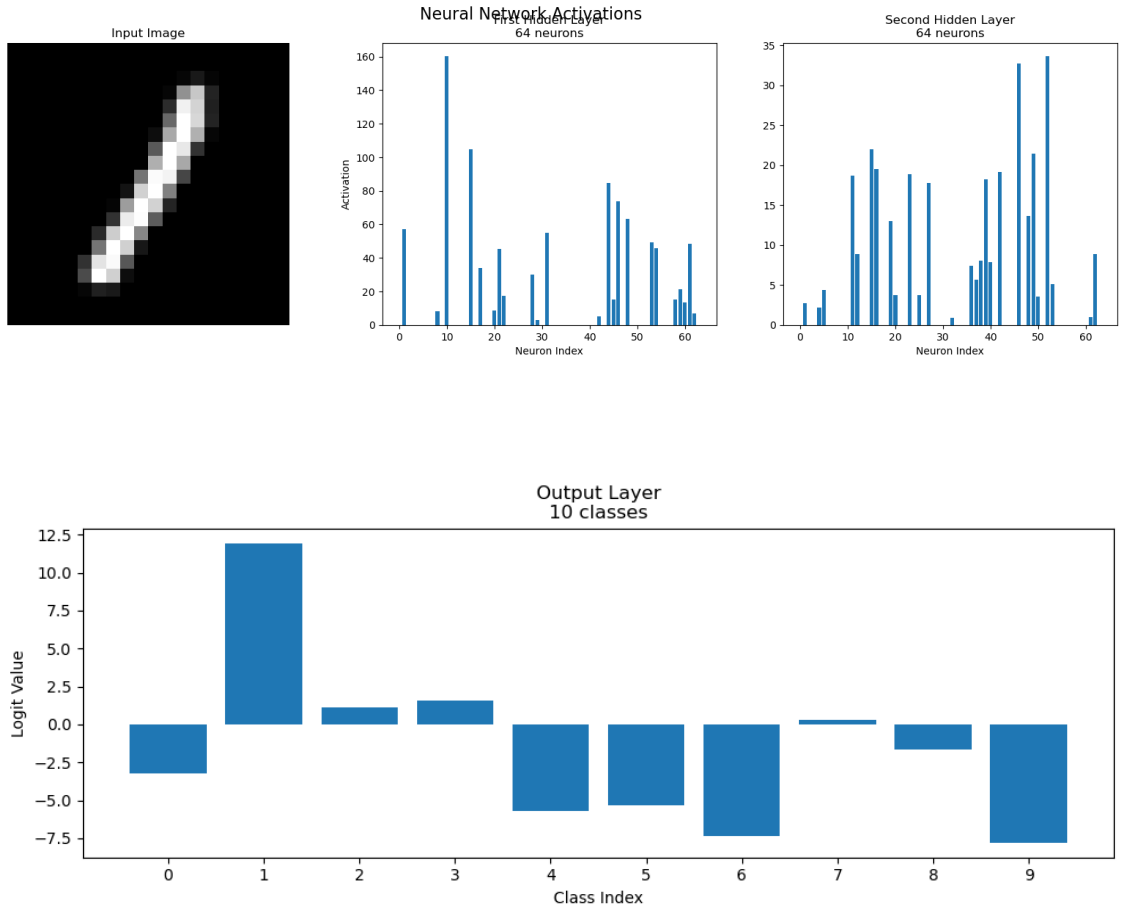
1.3 Activations

The neural network we have trained can now transform the input image into successive "activations" - the result of the linear neuron operation and the activation function, and the output at each layer. Let us visualize the image-dependent activations - not to be confused with visualizations of the (fixed) weights. Please read up about neural networks if you are unsure about activations or weights.

[12]:
```
def visualize_activations(model, input_image):
    """
    Visualize the activations at each layer of the network for a given input␣
↪image.

    Args:
        model: The PyTorch model
        input_image: A single input image (numpy array or tensor) of shape (1,␣
↪20, 20) or (400,)
    """
    if isinstance(input_image, np.ndarray):
        input_image = torch.from_numpy(input_image).float()

    # Ensure the input is 1D (flattened) and add batch dimension
    if input_image.dim() == 2:
        input_image = input_image.view(-1)
    if input_image.dim() == 1:
        input_image = input_image.unsqueeze(0)
    input_image = input_image.to(device)

    # Dictionary and hook function to store activations
    activations = {}
    def get_activation(name):
        def hook(model, input, output):
            activations[name] = output.detach()
        return hook

    # Register hooks for each layer
    hooks = []
    for i, layer in enumerate(model.linear_relu_stack):
        if isinstance(layer, nn.Linear) or isinstance(layer, nn.ReLU):
            hooks.append(layer.
↪register_forward_hook(get_activation(f'layer_{i}')))

    # Forward pass
    with torch.no_grad():
        model.eval()
        output = model(input_image)

    # Remove hooks
    for hook in hooks:
        hook.remove()

    fig, axes = plt.subplots(1, 3, figsize=(20, 5))
    fig.suptitle('Neural Network Activations', fontsize=16)

    if input_image.shape[1] == 400:  # If input is flattened
```

```
        img = input_image[0].cpu().numpy().reshape(20, 20)
    else:
        img = input_image[0].cpu().numpy()
    axes[0].imshow(img, cmap='gray')
    axes[0].set_title('Input Image')
    axes[0].axis('off')

    # Plot first hidden layer activations (after first Linear + ReLU)
    if 'layer_0' in activations and 'layer_1' in activations:
        act = activations['layer_1'].cpu().numpy()[0]  # Get first sample in␣
 ↪batch
        axes[1].bar(range(len(act)), act)
        axes[1].set_title(f'First Hidden Layer\n{len(act)} neurons')
        axes[1].set_xlabel('Neuron Index')
        axes[1].set_ylabel('Activation')

    # Plot second hidden layer activations (after second Linear + ReLU)
    if 'layer_2' in activations and 'layer_3' in activations:
        act = activations['layer_3'].cpu().numpy()[0]
        axes[2].bar(range(len(act)), act)
        axes[2].set_title(f'Second Hidden Layer\n{len(act)} neurons')
        axes[2].set_xlabel('Neuron Index')

    # Output layer activations (after final Linear)
    plt.figure(figsize=(10, 4))
    output_act = output.cpu().numpy()[0]
    plt.bar(range(len(output_act)), output_act)
    plt.title(f'Output Layer\n{len(output_act)} classes')
    plt.xlabel('Class Index')
    plt.ylabel('Logit Value')

    # Add class labels if available (assuming MNIST digits 0-9)
    plt.xticks(range(10), [str(i) for i in range(10)])

    plt.tight_layout()
    plt.show()

    # Print the predicted class
    predicted_class = torch.argmax(output, dim=1).item()
    print(f"Predicted class: {predicted_class}")
    print(f"Output logits: {output.cpu().numpy()[0].round(2)}")

# Example usage:
# Get a random test image
sample_idx = 250  # change this to see different examples, remember 250 samples␣
 ↪per digit
sample_image = test_tensor[sample_idx].unsqueeze(0)  # Add batch dimension
```

```
# Visualize activations
visualize_activations(model, sample_image)
```



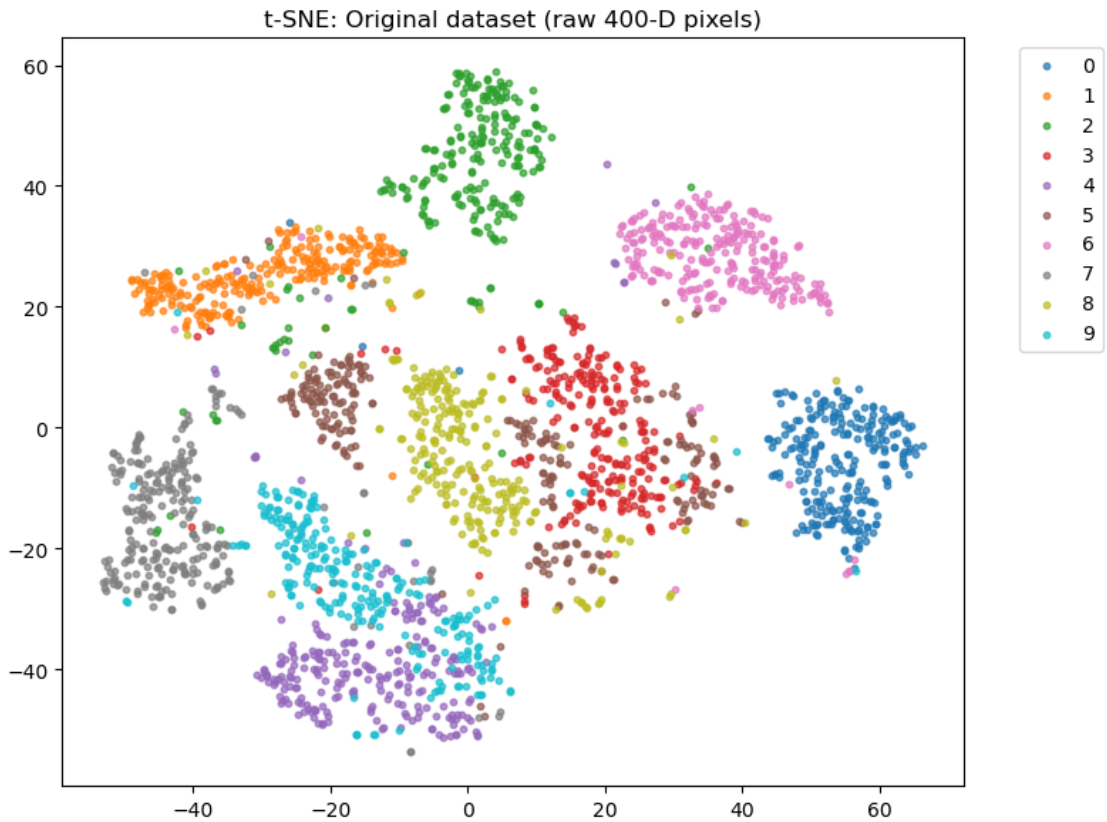

```
Predicted class: 1
Output logits: [-3.26 11.93  1.09  1.54 -5.74 -5.36 -7.32  0.32 -1.65 -7.78]
```

1.4 Lab: visualize the activation spaces

1. Visualize the first (0-th) layer activations of all training samples with t-SNE
2. Visualize the second layer activations with t-SNE
3. Visualize the third (output) layer activations with t-SNE
4. Qualitatively compare the t-SNE visualizations of the original dataset (from classification.ipynb) to the three visualizations of activations.

[15]:
```
x_raw = train #2500,400
y_raw = train_labels.squeeze() #2500x1
```

```
z_raw =␣
↪TSNE(n_components=2,perplexity=30,init="pca",learning_rate="auto",random_state=0).
↪fit_transform(x_raw)

plt.figure(figsize=(8,6))
for d in range(10):
    idx = (y_raw == d)
    plt.scatter(z_raw[idx,0], z_raw[idx,1], s=10, alpha=0.7, label=str(d))
plt.title("t-SNE: Original dataset (raw 400-D pixels)")
plt.legend(bbox_to_anchor=(1.05,1), loc="upper left")
plt.tight_layout()
plt.show()
```



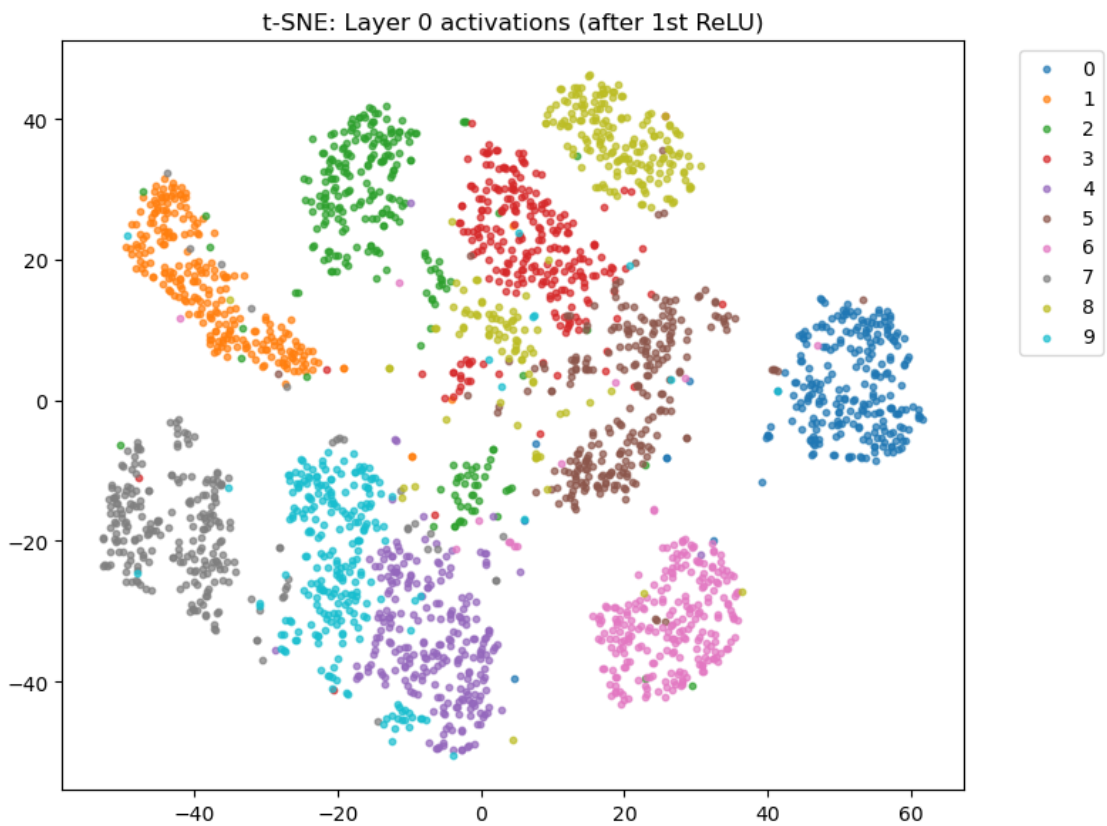
[ ]:
```
import numpy as np
import torch

model.eval() #-> NeuralNetwork (self)

A0_list, A1_list, A2_list, y_list = [], [], [], []
```

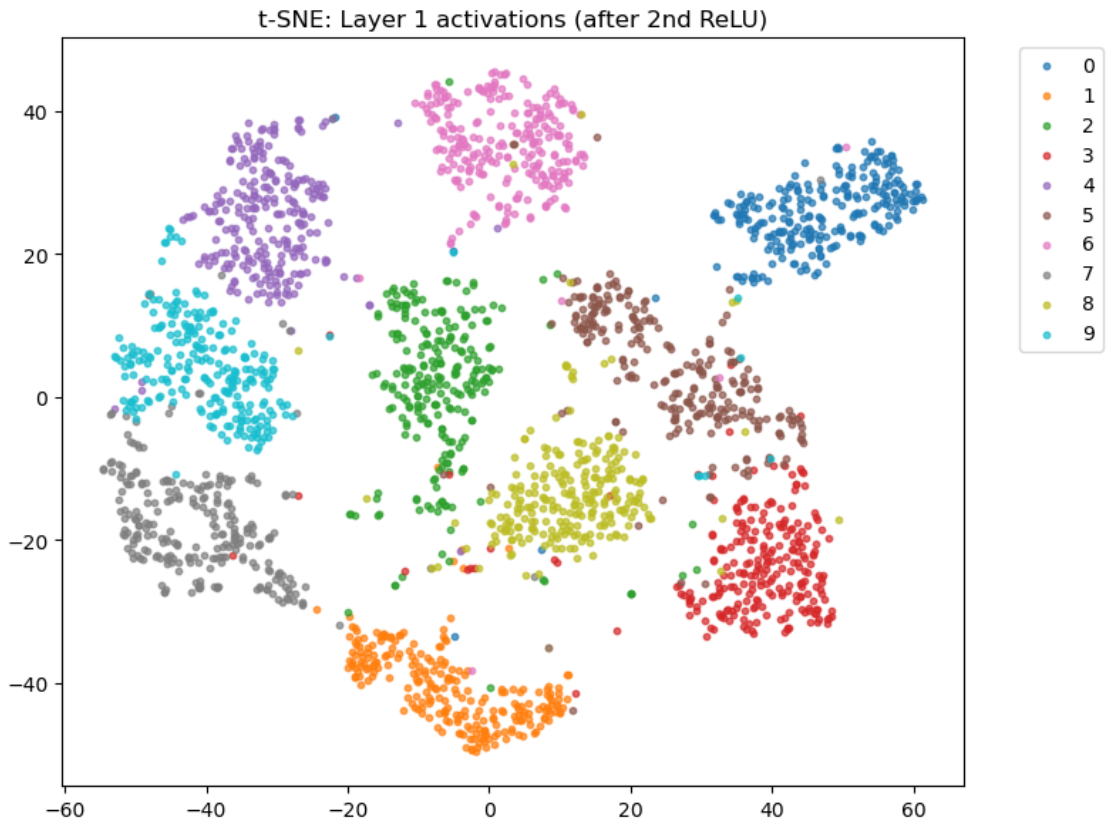
```
with torch.no_grad():
    '''torch.no_grad() Context-manager that disables gradient calculation.
    Disabling gradient calculation is useful for inference, when you are sure␣
 ↪that you will not call Tensor.backward().
    It will reduce memory consumption for computations that would otherwise␣
 ↪have requires_grad=True.'''
    for X, y in train_loader:
        X = X.to(device)

        x = model.flatten(X)                    #(B, 400)
        z0 = model.linear_relu_stack[0](x)      #Linear 400->64
        a0 = model.linear_relu_stack[1](z0)     #ReLU  (Layer 0 activations)
        z1 = model.linear_relu_stack[2](a0)     #Linear 64->64
        a1 = model.linear_relu_stack[3](z1)     # ReLU  (Layer 1 activation␣
 ↪application)

        a2 = model.linear_relu_stack[4](a1)     #Linear 64->10 logits (Output␣
 ↪layer - z)

        A0_list.append(a0.cpu().numpy())
        A1_list.append(a1.cpu().numpy())
        A2_list.append(a2.cpu().numpy())
        y_list.append(y.numpy())

A0 = np.vstack(A0_list)    #(N, 64)
A1 = np.vstack(A1_list)    #(N, 64)
A2 = np.vstack(A2_list)    #(N, 10)
y_train = np.concatenate(y_list)

print(A0.shape, A1.shape, A2.shape, y_train.shape)
```

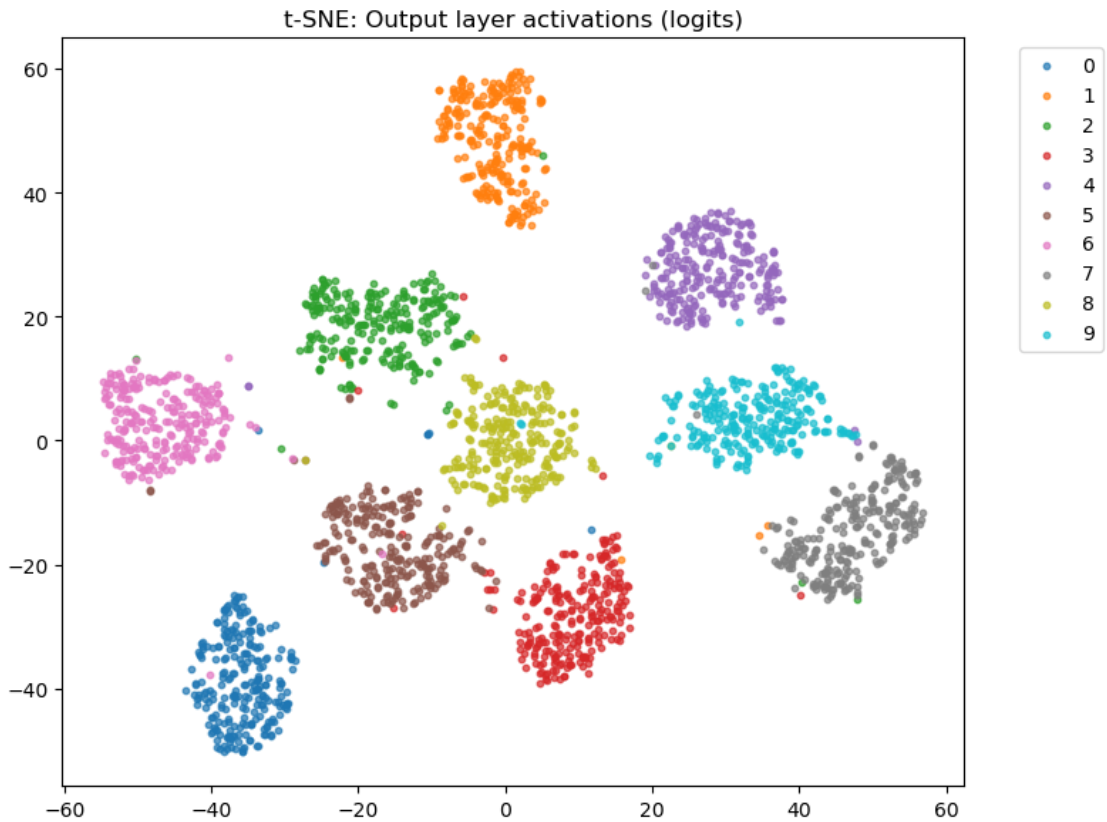
```
[[  0.        65.09612    0.        … 127.3849    12.717102   0.      ]
 [  0.        90.764465   0.        … 138.06177    0.         0.      ]
 [  0.         0.         0.        …  62.121643   0.         0.      ]
 …
 [  0.         0.         5.855633 …  60.829414   5.33309    0.      ]
 [  0.        52.327587   0.        …  44.860237   0.         0.      ]
 [  0.        17.259754  68.855255 …  34.899914  35.954376   0.      ]]
(2500, 64) (2500, 64) (2500, 10) (2500,)
```

[ ]:
```
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt

def tsne_plot(X, y, title,perplexity=30,seed=0):
    Z = TSNE(n_components=2, perplexity=perplexity, init="pca",␣
 ↪learning_rate="auto", random_state=seed).fit_transform(X)
    plt.figure(figsize=(8,6))
```

```
    for d in range(10):
        idx = (y == d)
        plt.scatter(Z[idx,0], Z[idx,1], s=10, alpha=0.7, label=str(d))
    plt.title(title)
    plt.legend(bbox_to_anchor=(1.05,1), loc="upper left")
    plt.tight_layout()
    plt.show()

#1) first (0-th) layer activations
tsne_plot(A0, y_train, "t-SNE: Layer 0 activations (after 1st ReLU)")
#2) second layer activations
tsne_plot(A1, y_train, "t-SNE: Layer 1 activations (after 2nd ReLU)")
#3) third (output) layer activations
tsne_plot(A2, y_train, "t-SNE: Output layer activations (logits)")
```




t-SNE: Layer 1 activations (after 2nd ReLU)
40
20
0
−20
−40
−60
−40
−20
0
20
40
60
0
1
2
3
4
5
6
7
8
9



[ ]:

-In the raw t-SNE plot, most of the digit classes blend into each other, which makes sense b/c pixel space doesn't separate them cleanly at all. -After the first hidden layer, you can start to see some structure. The digits form loose groups, though there's still plenty of mixing. -By the second hidden layer, the clusters become much tighter and more distinct. This is where the network seems to learn the most meaningful features. -At the output layer, the activations fall into very compact groups that line up with the final class predictions, showing how the model is carving out its decision boundaries.